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- Chi-squared and t-Distributions, Confidence Intervals, and Likelihoods

Confidence Intervals

All About Plotting

Bootstrap Principle, Algorithm, and Calculations

Review

Practice Quiz: Module 3 Homework

Quiz: Module 3 Quiz: Confidence Intervals, Bootstrapping, and Plotting Submitted

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Module 3 Quiz: Confidence Intervals, Bootstrapping, and Plotting

1. In a new population, a sample of 9 men yielded a sample average brain volume of 1,100cc and a standard deviation of 30cc. What is a 90% confidence interval for the mean brain volume in this new population?

Ready to review what you've learned before taking the quiz? I'm here to help.

🕒 1 / 1 point

📄

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2. A diet pill is given to 9 subjects. The average weight (follow up - baseline) is -2 pounds. What would the standard deviation have to be for the 95% T confidence interval to be entirely below 0?

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3. Refer to the previous question. The interval would up being [-3.5, -0.5] pounds. What can be said about the population mean weight for this population?

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4. In an effort to improve efficiency, hospital administrators are evaluating a new triage system for their emergency room. In an validation study of the system, 5 patients were tracked in a mock ER under both the new and old triage system. Their waiting times were recorded. Would it be better to use an independent group or paired T confidence interval in this setting?

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👍 Correct

5. Refer to the setting of the previous question. To further test the system, administrators selected 20 nights and randomly assigned the new triage system to be used on 10 nights and the standard system on the remaining 10 nights. They calculated the nightly median waiting time (MWT) to see a physician. The average MWT for the new system was 3 hours with a variance of 0.60 while the average MWT for the old system was 5 hours with a variance of 0.68. Give a 95% confidence interval estimate for the differences of the mean MWT associated with the new system. Assume a constant variance.

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👍 Correct

6. Suppose that you create a 95% T confidence interval. You then create a 90% interval using the same data. What can be said about the 90% interval with respect to the 95% interval?

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👍 Correct

7. Let distribution 1 be $N(\mu_1, \sigma_1^2)$ and distribution 2 be $N(\mu_2, \sigma_2^2)$. Let $x_{1-\alpha}$ and $x_{2-\alpha}$ be the α^{th} quantile from the two distributions, respectively. How are the two mathematically related?

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👍 Correct

8. Consider data points $x_1, \dots, x_n$. Imagine a probability mass function, so that a random variable X from this distribution has $P(X = x_i) = p(x_i) = \frac{1}{n}$. This is the so-called bootstrap distribution. What is the mean of the bootstrap distribution?

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👍 Correct

9. Suppose we were to simulate a large number of standard normal random variables and a large number of exponential random variables. What would a plot of the exponential quantiles (horizontal axis) versus the standard normal quantiles (vertical axis) look like? Let Φ be the standard normal distribution function.

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👍 Correct

10. Let $F(x)$ be a distribution function. Notice that $G(x) = F(a^2x + b)$ is also a distribution function for $a \neq 0$. If you were to take large samples from F and G , what must the QQ plot look like without knowing the specific values of a and b ?

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👍 Correct

11. Let your data be the two points $\{1, 3\}$. What is the bootstrap distribution of the sample mean?

🕒 1 / 1 point

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👍 Correct